

Pull-Up Ratio (PR) Derivation for 6T SRAM Write Stability

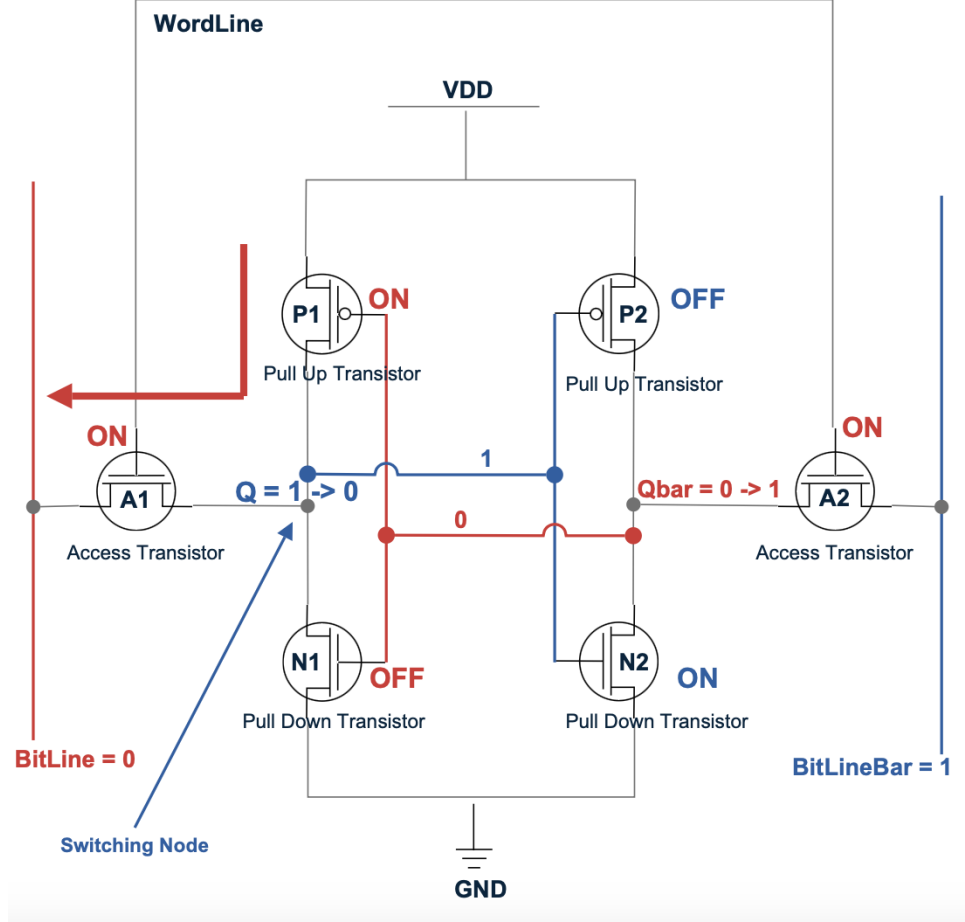


Figure 1: Critical switching instant during the Write-'0' operation of a 6T SRAM cell. Initially, the cell stores $Q = 1$ and $\bar{Q} = 0$. After asserting $WL = 1$, the Access NMOS (A1) attempts to discharge node Q through $BL = 0$, while the Pull-Up PMOS (P1) simultaneously tries to hold the same node at V_{DD} . Successful write operation requires the Access transistor to overpower the Pull-Up transistor, ensuring that node Q crosses the inverter switching threshold and the cell flips to the new state.

Write Stability Analysis

Assume that the SRAM cell initially stores

$$Q = 1, \quad \bar{Q} = 0$$

To write logic '0' into node Q ,

$$WL = V_{DD}, \quad BL = 0, \quad BLB = V_{DD}$$

Let the storage node voltage decrease to an intermediate value V_1 during the critical switching interval.

Access NMOS (A1)

$$V_G = V_{DD}, \quad V_D = V_1, \quad V_S = 0$$

Therefore,

$$V_{GS} = V_{DD}, \quad V_{DS} = V_1$$

Since

$$V_{DS} < V_{GS} - V_{TN},$$

the Access transistor operates in the linear region.

$$I_{A1} = \mu_n C_{ox} \left(\frac{W}{L} \right)_{A1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right]$$

Pull-Up PMOS (P1)

The Pull-Up PMOS remains ON during the critical switching interval.

$$V_S = V_{DD}, \quad V_G = 0, \quad V_D = V_1$$

Thus,

$$V_{SG} = V_{DD}, \quad V_{SD} = V_{DD} - V_1$$

Assuming

$$V_{SD} \geq V_{SG} - |V_{TP}|,$$

the Pull-Up transistor operates in saturation.

$$I_{P1} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{P1} (V_{DD} - |V_{TP}|)^2$$

Applying Kirchhoff's Current Law

At the critical storage node,

$$I_{A1} = I_{P1}$$

Substituting the drain current equations,

$$\mu_n C_{ox} \left(\frac{W}{L} \right)_{A1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right] = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_{P1} (V_{DD} - |V_{TP}|)^2$$

Cancelling the common term C_{ox} ,

$$\mu_n \left(\frac{W}{L} \right)_{A1} \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right] = \frac{1}{2} \mu_p \left(\frac{W}{L} \right)_{P1} (V_{DD} - |V_{TP}|)^2$$

Rearranging,

$$\boxed{\frac{\left(\frac{W}{L} \right)_{P1}}{\left(\frac{W}{L} \right)_{A1}} = \frac{2\mu_n \left[(V_{DD} - V_{TN})V_1 - \frac{V_1^2}{2} \right]}{\mu_p (V_{DD} - |V_{TP}|)^2}}$$

Physical Interpretation

During a Write-'0' operation, the Access NMOS transistor attempts to pull the storage node Q towards ground through the discharged bitline ($BL = 0$). Simultaneously, the Pull-Up PMOS tries to maintain the same node at V_{DD} . Consequently, these two transistors compete to determine the voltage of the storage node during the critical switching interval. If the Pull-Up transistor is excessively strong, the Access transistor cannot discharge node Q below the inverter switching threshold, preventing the cross-coupled inverters from changing state. Therefore, the Access transistor must be stronger than the Pull-Up transistor to ensure successful write operation.

Hence,

$$\boxed{\frac{(W/L)_{P1}}{(W/L)_{A1}} < 1}$$

or equivalently,

$$\boxed{PR = \frac{(W/L)_{P1}}{(W/L)_{A1}} < 1}$$

In practical SRAM designs,

$$\boxed{PR \approx 0.7 - 1}$$

is generally selected to provide sufficient write ability while maintaining acceptable cell stability.